



AI-Driven Weather Forecasting for Agriculture

"From Guessing the Rain to Predicting the Future 🌥️🤖"

Imagine a farmer checking the sky like a WhatsApp status: *"Clouds seen at 3 PM... rain expected? 🤔"* Now imagine AI saying: **"Rain at 4:12 PM. Carry umbrella. Also, don't plant today."** 👉 That's the power of AI.

☁️ The Problem with Traditional Forecasting

Inaccurate

Forecasts are generalised — not precise at village or farm level.

Delayed

Farmers rely on experience, guesswork, and "my grandfather said..." logic 😊

Costly Impact

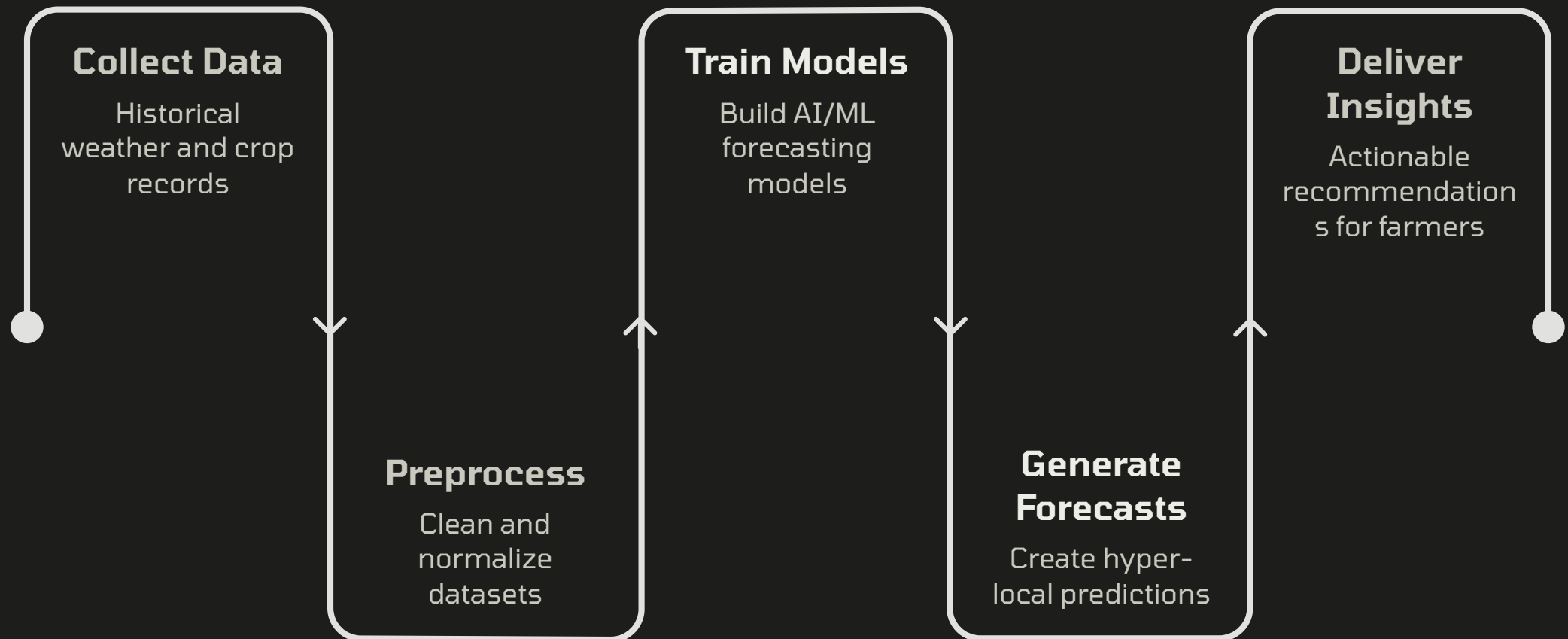
Crop failure, water wastage, and financial loss.

📺 Traditional forecasting: *"There might be rain somewhere in the state."* Farmer hears: **"Should I plant or not???"** 😭



Methodology: How It Works

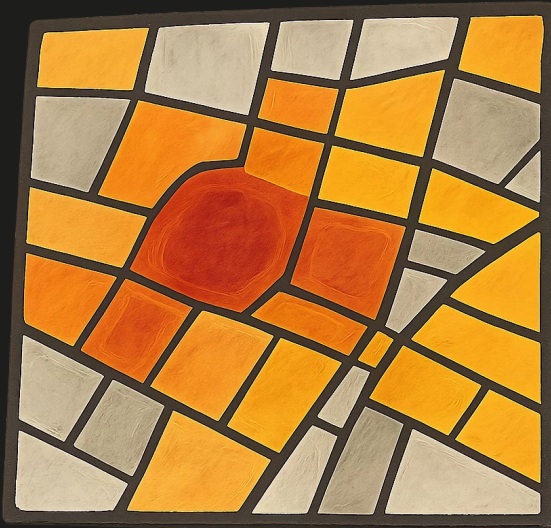
Think of AI as a super-smart farmer's assistant — never sleeps 🤖❌, reads millions of weather records 📊, and remembers everything (unlike humans 😊).



This end-to-end pipeline transforms raw environmental data into actionable, farm-level guidance for every grower.



Data Description



Data Sources

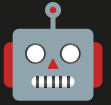
- Weather data — temperature, rainfall, humidity
- Soil data — moisture, nutrients
- Satellite imagery
- Historical crop yield data

Data Characteristics

- Time-series & spatial (location-based)
- High variability



🍲 Data is like cooking ingredients — too little = useless dish, too much = confusion, right mix = ★ Michelin-star prediction.



Models & Approach



Machine Learning

Random Forest & Support Vector Machines for pattern recognition.



Deep Learning

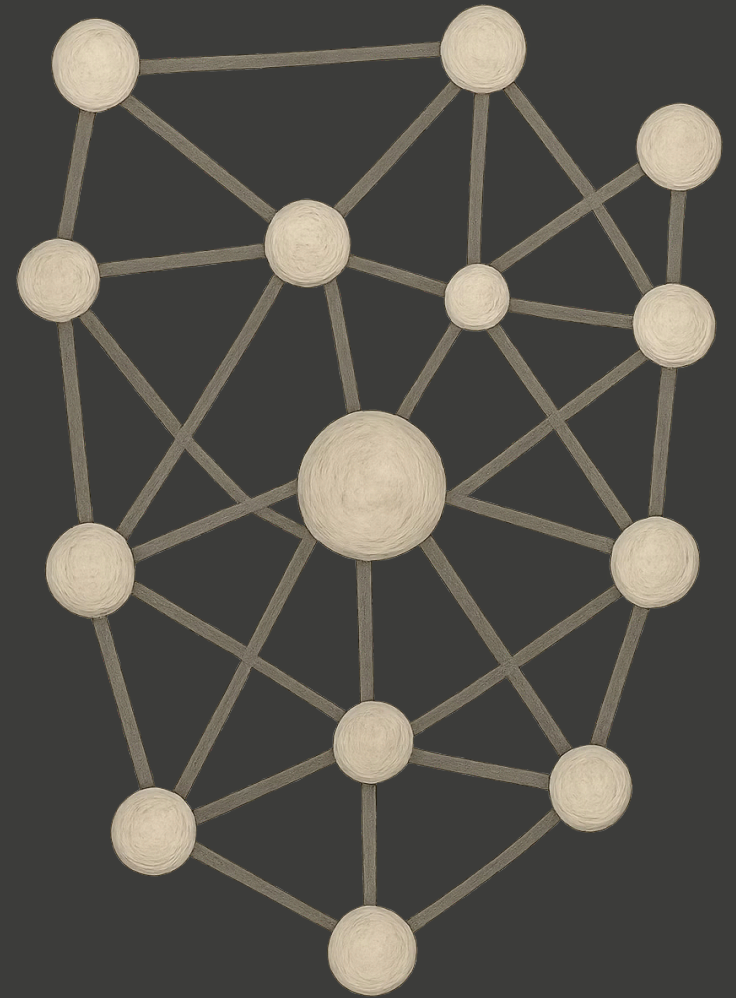
LSTM networks for time-series forecasting — because weather is like a Netflix series 📺 What happened yesterday affects today's episode.



Hybrid Models

AI + Satellite + IoT sensors for hyper-local precision.

Simple Flow: Input → Model → Prediction → Farmer Advice





Results & Real Impact

85–95%

Forecast Accuracy

Improved prediction rates with AI models.

What Farmers Gain

- Decide optimal sowing time
- Optimize water & irrigation use
- Avoid extreme weather risks
- Reduced crop loss & increased yield

❏ Before AI: *"Let's plant and pray 🙏"* After AI: **"Let's plant... based on data 😎📊"**



Conclusion: Smart Farmers 2.0



Sustainability

Less water waste, smarter resource use.



Profitability

Higher yields, reduced crop loss.



Efficiency

Data-driven decisions replace guesswork.

Future Scope: Real-time mobile alerts, voice assistants in local languages, integration with drones & IoT.

AI doesn't replace farmers...  It **upgrades** them to "**Smart Farmers 2.0**" 



? Questions & Discussion

Thank you! Now you can ask questions...

Or if you're a farmer, I can predict whether your question will be difficult 😊🌧️